

Artificial Intelligence

Homework 8

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November 11, 2025

Deadline

November 18, 2025

Pretraining language models - 80 Points

Training data

The training data contains the following 8 sentences:

1. I teach Computer Science in the Department of Computer Science at University of Alabama at Birmingham.
2. I am a Professor teaching Artificial Intelligence in the Department of Computer Science at University of Alabama at Birmingham.
3. I am a student at University of Alabama at Birmingham.
4. I study computer science as a student at University of Alabama at Birmingham.
5. A professor and a student are at University of Alabama at Birmingham.
6. She is a nurse working in the School of Medicine.
7. He is a doctor working in the School of Medicine.
8. A doctor and a nurse work in the School of Medicine.

Testing data

The testing data contains the following 6 sentences with a missing / masked word that is denoted by [MASK] token. Your task is to build a pretrained language model in order to predict the missing / masked word.

1. In the Department of Computer Science at University of Alabama at Birmingham, I teach Computer Science and Artificial Intelligence and I am a [MASK].
2. At University of Alabama at Birmingham, I study computer science and I am a [MASK].
3. We are at the University of Alabama at [MASK].
4. In the School of Medicine, he works as a [MASK].
5. In the School of Medicine, she works as a [MASK].
6. They work in the School of [MASK].

The training and testing data are included in the file `Homework.8.txt` in the folder HW on Canvas.

Task Description

In this assignment, you will **build and pretrain** neural language models in PyTorch on this small dataset:

- **Data pre-processing:** You have to preprocess the data (both training and testing) and create a vocabulary of unique words. You have to transform all words into lower-case. The input word embedding can be simply **one-hot encoding**. That means if the vocabulary contains V unique words and a sentence has N words then the input tensor is of size $N \times V$. For the [MASK] token, the corresponding embedding is a vector of length V of all zeros.
- The pretraining task is **Masked Language Models (MLM)**. You can mask 1, 2, 3, or 4 words randomly from a sentence. You have to create a pretraining data of at least 100 examples.
- You have to build 2 models:
 1. One model based on **Long Short-Term Memory (LSTM)**
 2. One model based on **Transformer**

You can use the builtin modules of PyTorch. You have the freedom to design the models as you wish as long as it uses LSTM and Transformer. You have to write a report about your training procedure including the loss function, the metrics, number of epochs, performance on each epoch, etc.

- After pretraining, you have to use your models to **predict the missing / masked word** in the test dataset. You have to report the testing results including the predicted words and overall accuracy.

Remember to submit all your codes and the report.

Using LLM API and chatbot - 20 Points

In addition, you can creatively use **LLM API (10 Points)** and **Gemini chatbot (or other) (10 Points)** to predict the missing words. You have to submit your code as well as your prompts and conversation with the chatbot.

Note: For humans, it is quite trivial to get the correct predictions on the testing set. I expect LLM that was pretrained on a vast amount of data could correctly predict all the missing / masked words. However, in order for LLM to make the correct predictions, you have to “adapt” it to the new training data. In practice, this “adaptation” is called the **fine-tuning phase**.